

11.1 Generative Adversarial Nets

A generator fakes samples. A discriminator tries to tell them from the data.

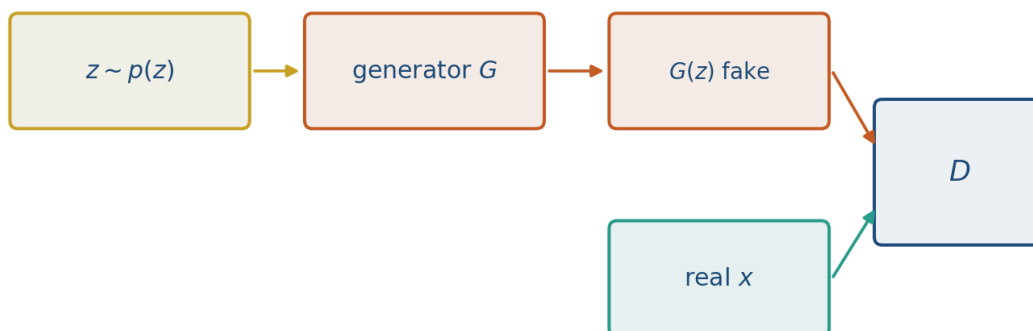
These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

A **GAN** (Goodfellow et al., 2014) is two networks in a contest: a generator that paints fakes, and a discriminator that tries to catch them. You do not write down $p(x)$ and maximize likelihood. You train a sampler.

1. The generator

Noise z (usually isotropic Gaussian) in; a sample $x = G(z; \theta_g)$ out, same shape as a real observation. G is a decoder without an encoder: it does not reconstruct a particular training point. It maps latent draws into data space.

GAN: G tries to fool D ; D tries not to be fooled



The job of G is to make the **distribution** of fakes look like the distribution of reals. One pretty sample is not a trained generator.

2. The discriminator

$D(x; \theta_d)$ is a binary classifier: the probability that x is real. On a real batch it should sit near 1; on $G(z)$ it should sit near 0. It never sees z . It only sees the sample.

$$y = D(x; \theta_d) \in (0, 1).$$

Then $1 - D(x)$ is the probability it calls the point fake.

3. Why two players

If you trained only G against a frozen D , G would overfit whatever D currently likes. If you trained only D , it would become a perfect detector of yesterday's fakes and give G a useless (flat) gradient. The pair has to stay in tension. Note **11.2** writes that as a min-max game; note **11.3** is what happens when G finds one easy fake and stops covering the rest.

For this course, GANs are the first **implicit** generator you meet. Diffusion (Week 12) will give you an explicit noise schedule instead of an adversary. Text-to-image later in the module still needs a way to **condition** (Week 5's CLIP is one).

4. Practice

1. G never sees a real image on its own update (only through D 's scores). How can it learn the data distribution?
2. Why is a perfect D (always 1 on reals, always 0 on fakes) a problem for G 's gradient?
3. Name one task where a GAN is a natural fit and one where an explicit likelihood model would be easier to evaluate.